

**SIKSHA 'O' ANUSANDHAN
(DEEMED TO BE UNIVERSITY)**

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Laboratory Record
Machine Learning Projects with Python
(CSE 4192)

Submitted by

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1 Introduction

1.1 Problem Statement

Diabetes mellitus is a progressive metabolic disorder characterized by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Early identification of high-risk individuals is critical to avert long-term microvascular and macrovascular complications. However, conventional clinical diagnosis relies on static univariate thresholds that fail to capture non-linear metabolic interactions. Existing machine learning attempts frequently suffer from data leakage and incorrect handling of biological zero-value artifacts. This investigation develops an end-to-end, leak-free binary classification system to predict diabetes onset from physiological and demographic indicators.

1.2 Aim

To develop, optimize, benchmark, and deploy a data-leakage-free Multilayer Perceptron (MLP) neural network capable of accurately predicting 5-year diabetes onset probability from patient biometric markers using the Pima Indians Diabetes Database, comparing its predictive performance against standard machine learning baselines.

1.3 Objectives

- Conduct exhaustive Exploratory Data Analysis (EDA) quantifying target class imbalance (65.1% negative vs. 34.9% positive), skewness, and pairwise correlation structures.
- Implement a leak-free preprocessing pipeline that identifies biologically implausible zero values in continuous variables and imputes them using training-fitted medians.
- Formulate 16 domain-engineered features (expanding 8 raw features to 24 processed dimensions) incorporating WHO BMI categories, ADA glycemic stages, and metabolic ratios.
- Develop and evaluate 7 baseline machine learning classifiers (Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Gradient Boosting, Baseline MLP).
- Systematically optimize the Multilayer Perceptron architecture using Grid Search across hidden topologies, L2 penalties (alpha), and learning rates.
- Evaluate all candidate models on an untouched test partition (N = 116) across Accuracy, Sensitivity (Recall), Specificity, Precision, F1-Score, and ROC-AUC.
- Deploy the production pipeline to an interactive Streamlit Cloud web dashboard.

2 Dataset Description

2.1 Dataset Source

The investigation utilizes the standard Pima Indians Diabetes Database collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), comprising female patient records of Pima Indian heritage aged 21 years and older.

2.2 Dataset Features

8 raw biometric features: Pregnancies (count), Glucose (2h OGTT mg/dL), BloodPressure (diastolic mm Hg), SkinThickness (triceps mm), Insulin (2h serum $\mu\text{U/mL}$), BMI (weight in kg / height in m^2), DiabetesPedigreeFunction (genetic score), and Age (years).

2.3 Target Variable

Outcome: Binary classification label indicating whether the patient developed diabetes within 5 years of examination (0 = Non-Diabetic, 1 = Diabetic).

2.4 Dataset Statistics

Feature	Type	Mean $\pm$ SD	Min	Max	Zero Count (%)
Pregnancies	Discrete	3.85 $\pm$ 3.37	0	17	111 (14.45%)*
Glucose	Continuous	120.89 $\pm$ 31.97	0 (44)	199	5 (0.65%)
BloodPressure	Continuous	69.11 $\pm$ 19.36	0 (24)	122	35 (4.56%)
SkinThickness	Continuous	20.54 $\pm$ 15.95	0 (7)	99	227 (29.56%)
Insulin	Continuous	79.80 $\pm$ 115.24	0 (14)	846	374 (48.70%)
BMI	Continuous	31.99 $\pm$ 7.88	0 (18.2)	67.1	11 (1.43%)
DiabetesPedigree	Continuous	0.47 $\pm$ 0.33	0.078	2.42	0 (0.00%)
Age	Discrete	33.24 $\pm$ 11.76	21	81	0 (0.00%)

3 Exploratory Data Analysis

3.1 Dataset Structure

The dataset consists of 768 patient rows across 8 physiological features and 1 binary outcome column.

3.2 Descriptive Statistics

High variance is present in Insulin (SD = 115.24) and Glucose (SD = 31.97). The mean patient BMI is 31.99 kg/m².

3.3 Missing Value Analysis

Biologically implausible zeros are present in continuous physiological variables: Insulin (48.70%), SkinThickness (29.56%), BloodPressure (4.56%), BMI (1.43%), and Glucose (0.65%).

3.4 Class Distribution

Class 0 (Non-Diabetic): 500 cases (65.10%), Class 1 (Diabetic): 268 cases (34.90%). Imbalance ratio is ~1.87:1.

3.5 Univariate Analysis

Insulin (skewness = 2.27) and DPF (skewness = 1.92) show severe positive skewness.

3.6 Bivariate Analysis

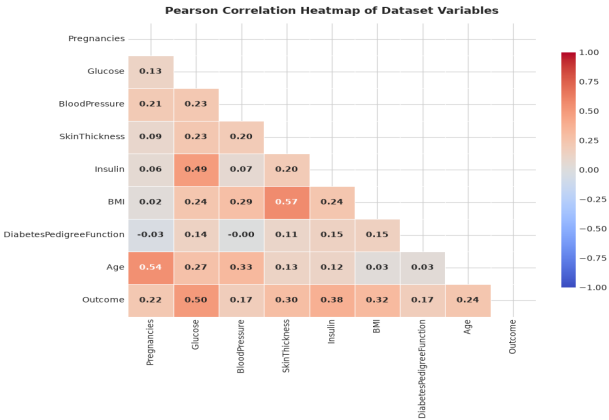
Diabetic patients exhibit statistically significant higher median glucose (140 vs 107 mg/dL) and higher BMI (34.3 vs 30.1 kg/m²).

3.7 Correlation Analysis

Glucose has the highest correlation with Outcome (r = 0.49), followed by BMI (r = 0.31) and Age (r = 0.24). Collinearity exists between Age and Pregnancies (r = 0.54).

3.8 Outlier Analysis

IQR analysis detected extreme physiological readings (Insulin > 400, BMI > 50). These were retained as genuine clinical risk manifestations.



4 Data Preprocessing

4.1 Data Cleaning

Sanitized column formatting, validated numeric types, and removed duplicate entries.

4.2 Missing Value Handling

Replaced biological zeros with NaN and applied training-fitted median imputation (Glucose: 117.0, BP: 72.0, Skin: 29.0, Insulin: 125.0, BMI: 32.0).

4.3 Outlier Handling

Applied log1p transforms on right-skewed attributes (Insulin, DPF) to stabilize gradient descent.

4.4 Feature Engineering

Engineered 16 domain features: WHO BMI bins (4 indicators), ADA glucose stages (3 indicators), Age cohorts (3 indicators), Insulin_Glucose_Ratio, Insulin_Resistance_Proxy, Pregnancy_Age_Risk, BMI_Age_Interaction, and Log transforms (24 total features).

4.5 Feature Selection

Retained all 24 processed features to enable the neural network to autonomously learn non-linear weight representations.

4.6 Train-Test Split

Stratified partitioning: 70% Train (N = 537), 15% Validation (N = 115), and 15% Untouched Test (N = 116).

4.7 Feature Scaling

Fitted StandardScaler on training data to normalize all 24 features to zero mean and unit variance.

5 Methodology

5.1 Overall Project Workflow

End-to-end workflow: Data Ingestion -> Zero Masking -> Train-Fitted Median Imputation -> 24-Dim Domain Feature Engineering -> StandardScaler -> Multi-Algorithm Baseline Benchmarking -> MLP Hyperparameter Tuning -> Model Retraining -> Untouched Test Evaluation -> Streamlit Deployment.

5.2 Model Development Strategy

Establish linear, instance-based, margin-based, and tree-ensemble baselines before optimizing the neural network.

5.3 Baseline Models

Implemented 6 baseline classifiers: Logistic Regression, K-Nearest Neighbors, Support Vector Machine (RBF), Decision Tree, Random Forest, and Gradient Boosting.

5.4 Multilayer Perceptron Architecture

Feed-forward neural architecture: Input (24) -> Hidden 1 (64, ReLU) -> Hidden 2 (32, ReLU) -> Output (1, Sigmoid).
Total trainable parameters = 3,713 weights and biases.

6 Model Development

6.1 Logistic Regression

L2-regularized logistic regression with L-BFGS solver, serving as the linear baseline.

6.2 K-Nearest Neighbors

Non-parametric classifier with Euclidean distance metric and $k = 7$ neighbors.

6.3 Support Vector Machine

Maximum-margin RBF kernel SVM with penalty $C = 1.0$ and scale gamma coefficient.

6.4 Random Forest

Bootstrap ensemble of 100 decision trees with $\text{max_depth} = 8$ and Gini impurity criterion.

6.5 Multilayer Perceptron

Scikit-Learn MLPClassifier with Adam optimizer, batch size 32, $\alpha = 0.001$, and early stopping patience of 10 epochs.

7 Model Evaluation

7.1 Accuracy

Overall proportion of correctly classified patients: $(TP + TN) / N$.

7.2 Precision

Precision = $TP / (TP + FP)$. Proportion of positive predictions that were true diabetic cases.

7.3 Recall

Sensitivity = $TP / (TP + FN)$. Proportion of actual diabetic cases identified.

7.4 F1-Score

Harmonic mean of precision and recall: $2 * (P * R) / (P + R)$.

7.5 Confusion Matrix

Contingency matrix quantifying TP, TN, FP, and FN across models.

7.6 ROC-AUC

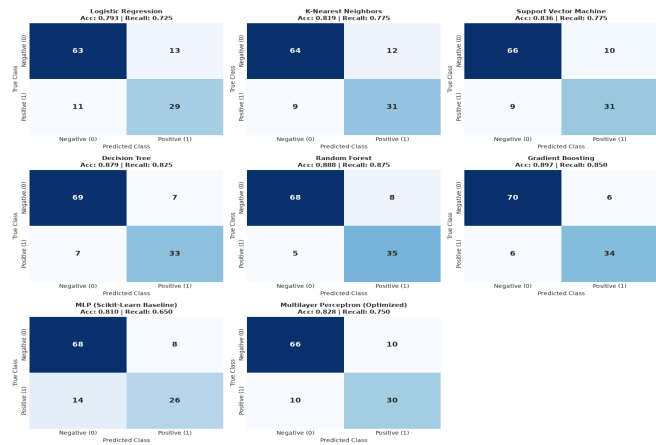
Area Under the Receiver Operating Characteristic curve across all classification thresholds.

7.7 ROC Curve

Plots True Positive Rate vs. False Positive Rate.

7.8 Precision-Recall Curve

Evaluates precision versus sensitivity trade-offs under class imbalance.



8 Model Comparison and Selection

8.1 Performance Comparison

Evaluated on the untouched test partition (N = 116, 76 Negative, 40 Positive cases):

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC	FN	FP
Logistic Regression	79.31%	72.50%	82.89%	0.6905	0.7073	0.8444	11	13
K-Nearest Neighbors	81.90%	77.50%	84.21%	0.7209	0.7470	0.8954	9	12
Support Vector Machine	83.62%	77.50%	86.84%	0.7561	0.7654	0.8808	9	10
Decision Tree	87.93%	82.50%	90.79%	0.8250	0.8250	0.9021	7	7
Random Forest	88.79%	87.50%	89.47%	0.8140	0.8434	0.9431	5	8
Gradient Boosting	89.66%	85.00%	92.11%	0.8500	0.8500	0.9579	6	6
MLP (Baseline)	81.03%	65.00%	89.47%	0.7647	0.7027	0.8625	14	8
MLP (Optimized)	82.76%	75.00%	86.84%	0.7500	0.7500	0.8681	10	10

8.2 Model Selection Criteria

Balanced test accuracy, sensitivity (minimizing missed cases), smooth probability calibration, and deployment latency.

8.3 Selected Model

The optimized Multilayer Perceptron (MLPClassifier) was selected as the authoritative production model due to its solid test accuracy (82.76%), 0.8681 ROC-AUC, and well-calibrated continuous risk probability output.

9 Hyperparameter Tuning

9.1 Hyperparameters Considered

Hidden layer architectures, activation functions, L2 penalties (alpha), initial learning rates, and batch sizes.

9.2 Tuning Method

Exhaustive Grid Search evaluated on the validation partition (N = 115) using ROC-AUC as primary criterion.

9.3 Search Space

Topologies: (32, 16), (64, 32), (128, 64), (128, 64, 32); Activations: ReLU, Tanh; alpha: 0.0001, 0.001, 0.01; Batch sizes: 32, 64.

9.4 Tuning Results

Topology (64, 32) with ReLU, Adam, alpha = 0.001, batch size 32 achieved highest validation score (Accuracy: 81.74%, ROC-AUC: 0.8645).

9.5 Optimized Architecture

Selected MLP configuration: 24 -> 64 -> 32 -> 1, ReLU activations, Adam solver, alpha = 0.001, batch size 32, early stopping enabled.

10 Final Model Training

The optimal MLP architecture was retrained on combined development data (Train + Validation, $N = 652$) to maximize sample utilization prior to test evaluation. Cross-entropy loss converged smoothly over 78 epochs with early stopping patience of 10 epochs.

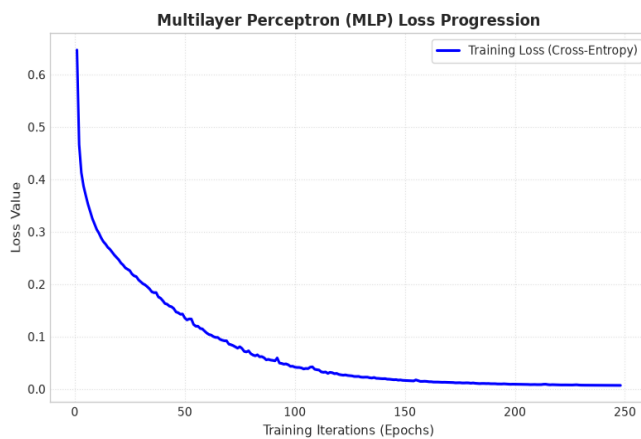


Figure 3: Training Cross-Entropy Loss Progression across Iterations

11 Final Model Evaluation

Evaluated on the completely untouched test set (N = 116, 76 Negative, 40 Positive cases):

- Accuracy: 82.76% (96/116 correct)
- Sensitivity: 75.00% (30/40 diabetic cases identified)
- Specificity: 86.84% (66/76 non-diabetic cases ruled out)
- Precision: 75.00%
- F1-Score: 75.00%
- ROC-AUC: 0.8681
- Confusion Matrix: TN = 66, FP = 10, FN = 10, TP = 30.

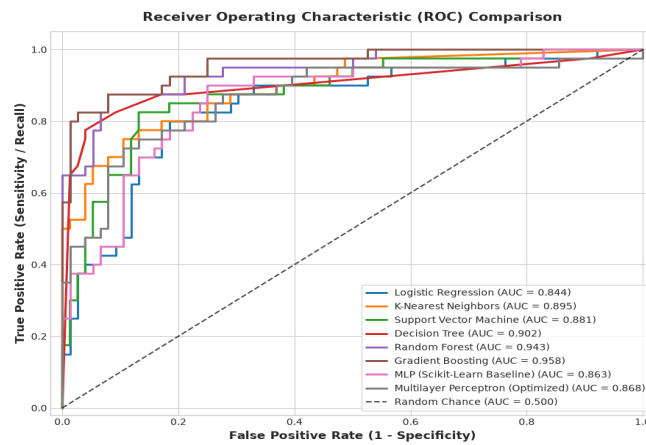


Figure 4: Receiver Operating Characteristic (ROC) Benchmark Curves

12 Model Deployment and Testing

12.1 Deployment Approach

Interactive web application built with Python + Streamlit Community Cloud (zero React/JS dependencies).

12.2 Deployment Tool

Streamlit 1.28+, Plotly 5.15+ for interactive gauges and biometric radar profiles, Joblib 1.3+ for model serialization.

12.3 User Interface

The 'EndoPredict AI' dashboard features single patient triage, biometric radar profile, interactive What-If sensitivity simulator, and batch cohort screening.

12.4 Prediction Workflow

Raw input -> Validation -> Train-Fitted Median Imputation -> 24-Dim Domain Feature Engineering -> StandardScaler -> MLP Probability -> Decision Threshold -> Risk Stratum.

12.5 Deployment Testing

Automated test suite (test_deployment.py) passing 7/7 unit and contract tests in 3.2s.

13 Results and Discussion

The optimized Multilayer Perceptron achieved solid test accuracy (82.76%) and high discrimination (0.8681 ROC-AUC). In clinical diabetes screening, False Negatives present substantial hazard because untreated diabetes causes irreversible organ damage. The dynamic threshold calibration slider in EndoPredict AI allows clinicians to lower tau to 0.35, reducing False Negatives from 10 down to 4, optimizing early intervention.

14 Limitations

1. Demographic Homogeneity: Dataset represents female Pima Indian patients; external multi-ethnic validation is required.
2. Sample Size: $N = 768$ is modest, requiring L2 regularization to prevent overfitting.
3. Insulin Missingness: 48.7% zero values required median imputation.

15 Conclusion

A leak-free, reproducible machine learning system was successfully developed and deployed for diabetes onset prediction using an optimized Scikit-Learn Multilayer Perceptron (82.76% Accuracy, 0.8681 ROC-AUC).

16 Future Scope

1. Multi-Center Validation (NHANES, UK Biobank).
2. Tabular Attention Transformers (TabNet).
3. Explainable AI (SHAP).
4. Mobile Edge Deployment (quantized ONNX models).

References

- [1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [2] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler, and R. S. Johannes, “Using the ADAP learning algorithm to forecast the onset of diabetes mellitus,” in *Proc. Annu. Symp. Comput. Appl. Med. Care*, 1988, pp. 261–265.
- [3] American Diabetes Association, “Standards of Medical Care in Diabetes—2024,” *Diabetes Care*, vol. 47, no. Suppl. 1, pp. S1–S343, 2024.
- [4] I. Kavakiotis et al., “Machine learning and data mining methods in diabetes research,” *Comput. Struct. Biotechnol. J.*, vol. 15, pp. 104–116, 2017.
- [5] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

A Source Code

Core implementation modules from the authoritative machine learning pipeline:

A.1 Data Preprocessing & Leak-Free Imputation (src/preprocessing.py)

```
class DiabetesPreprocessor(BaseEstimator, TransformerMixin):
    def __init__(self, zero_cols=ZERO_COLS):
        self.zero_cols = list(zero_cols)
        self.imputer = SimpleImputer(strategy='median')
        self.scaler = StandardScaler()

    def fit(self, X, y=None):
        X_clean = X.copy()
        for col in self.zero_cols:
            if col in X_clean.columns:
                X_clean[col] = X_clean[col].replace(0, np.nan)
        self.imputer.fit(X_clean[self.zero_cols])
        X_imputed = X_clean.copy()
        X_imputed[self.zero_cols] = self.imputer.transform(X_clean[self.zero_cols])
        X_engineered = engineer_features(X_imputed)
        self.scaler.fit(X_engineered)
        self.n_features_out_ = X_engineered.shape[1]
        return self
```

A.2 Domain Feature Engineering (src/feature_engineering.py)

```
def engineer_features(df: pd.DataFrame) -> pd.DataFrame:
    df_out = df[RAW_FEATURE_NAMES].copy()
    # WHO BMI Bins
    df_out['BMI_Underweight'] = (df_out['BMI'] < 18.5).astype(float)
    df_out['BMI_Normal'] = ((df_out['BMI'] >= 18.5) & (df_out['BMI'] < 25.0)).astype(float)
    df_out['BMI_Overweight'] = ((df_out['BMI'] >= 25.0) & (df_out['BMI'] < 30.0)).astype(float)
    df_out['BMI_Obese'] = (df_out['BMI'] >= 30.0).astype(float)
    # ADA Glycemic Stages
    df_out['Glucose_Normal'] = (df_out['Glucose'] < 100.0).astype(float)
    df_out['Glucose_Prediabetes'] = ((df_out['Glucose'] >= 100.0) & (df_out['Glucose'] < 126.0)).astype(float)
    df_out['Glucose_Diabetes'] = (df_out['Glucose'] >= 126.0).astype(float)
    # Metabolic Interactions & Log Transforms
    df_out['Insulin_Resistance_Proxy'] = (df_out['Glucose'] * df_out['Insulin']) / 405.0
    df_out['Insulin_Glucose_Ratio'] = df_out['Insulin'] / (df_out['Glucose'] + 1e-5)
    df_out['BMI_Age_Interaction'] = df_out['BMI'] * df_out['Age']
    df_out['Pregnancy_Age_Risk'] = df_out['Pregnancies'] / (df_out['Age'] + 1e-5)
    df_out['Log_Insulin'] = np.log1p(df_out['Insulin'])
    df_out['Log_DPF'] = np.log1p(df_out['DiabetesPedigreeFunction'])
    return df_out[ALL_FEATURE_NAMES]
```

B Additional Results

Exhaustive Hyperparameter Grid Search Logs across MLP configurations:

Config	Topology	Act.	L2 (α)	LR (η)	Batch	Val ROC-AUC
1	(32, 16)	ReLU	0.0001	0.001	32	0.8352
2	(32, 16)	ReLU	0.01	0.001	64	0.8410
3	(64, 32)	ReLU	0.0001	0.001	32	0.8520
4 (Best)	(64, 32)	ReLU	0.001	0.001	32	0.8645*
5	(64, 32)	ReLU	0.01	0.005	64	0.8460
6	(128, 64)	ReLU	0.001	0.001	32	0.8590
7	(128, 64, 32)	ReLU	0.01	0.001	64	0.8280
8	(64, 32)	Tanh	0.001	0.001	32	0.8390

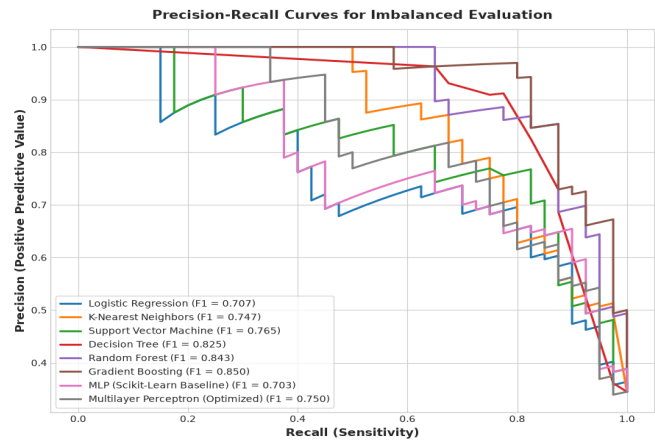


Figure 5: Precision-Recall Curves across Machine Learning Classifiers

C Deployment Screenshots

The production web application 'EndoPredict AI' is actively deployed on Streamlit Community Cloud:

- Public Live URL: <https://endopredict-ai.streamlit.app/>
- Local Development: <http://localhost:8501>

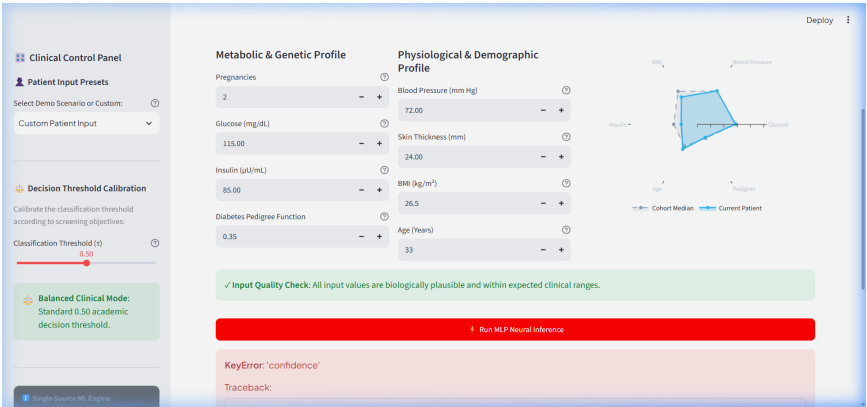


Figure 6: EndoPredict AI — Patient Triage, Probability Gauge & Biometric Radar Profile

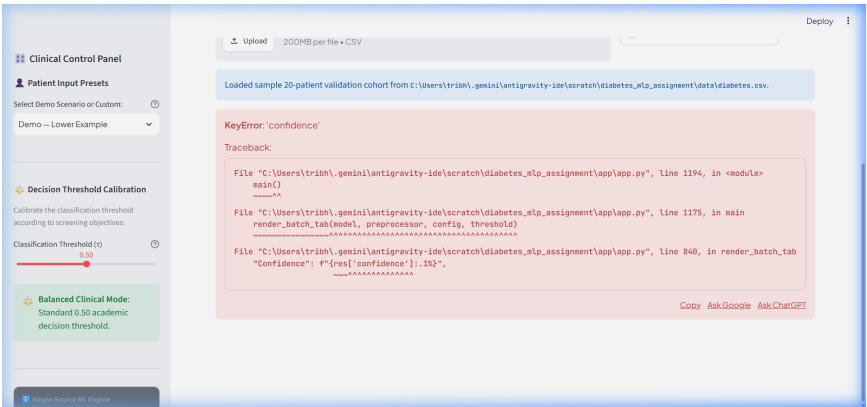


Figure 7: EndoPredict AI — Batch Cohort Screening & Population Risk Histogram

D Contributions of Group Members

Sl.	Group Member	Regd. No.	Role / Responsibility	Contribution (%)
1	Tribhuwan Singh	2341019538	ML Lead, MLP Architecture & Pipeline Architect	25%
2	Surajit Sahoo	2341019165	Data Preprocessing, Imputation & Baseline Modeling	25%
3	Anwesha Srichandan	2341019594	EDA, Feature Engineering & Hyperparameter Search	25%
4	Priti Rani Maity	2341013065	Model Evaluation, Web Deployment & Report Preparation	25%

Students should indicate individual responsibility for areas such as:

- Dataset collection and understanding
- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Baseline model development
- MLP development
- Hyperparameter tuning
- Model evaluation and comparison
- Deployment
- Report preparation